

ELECTRÓNICA DIGITAL III

CLASE 5

Módulo Timer

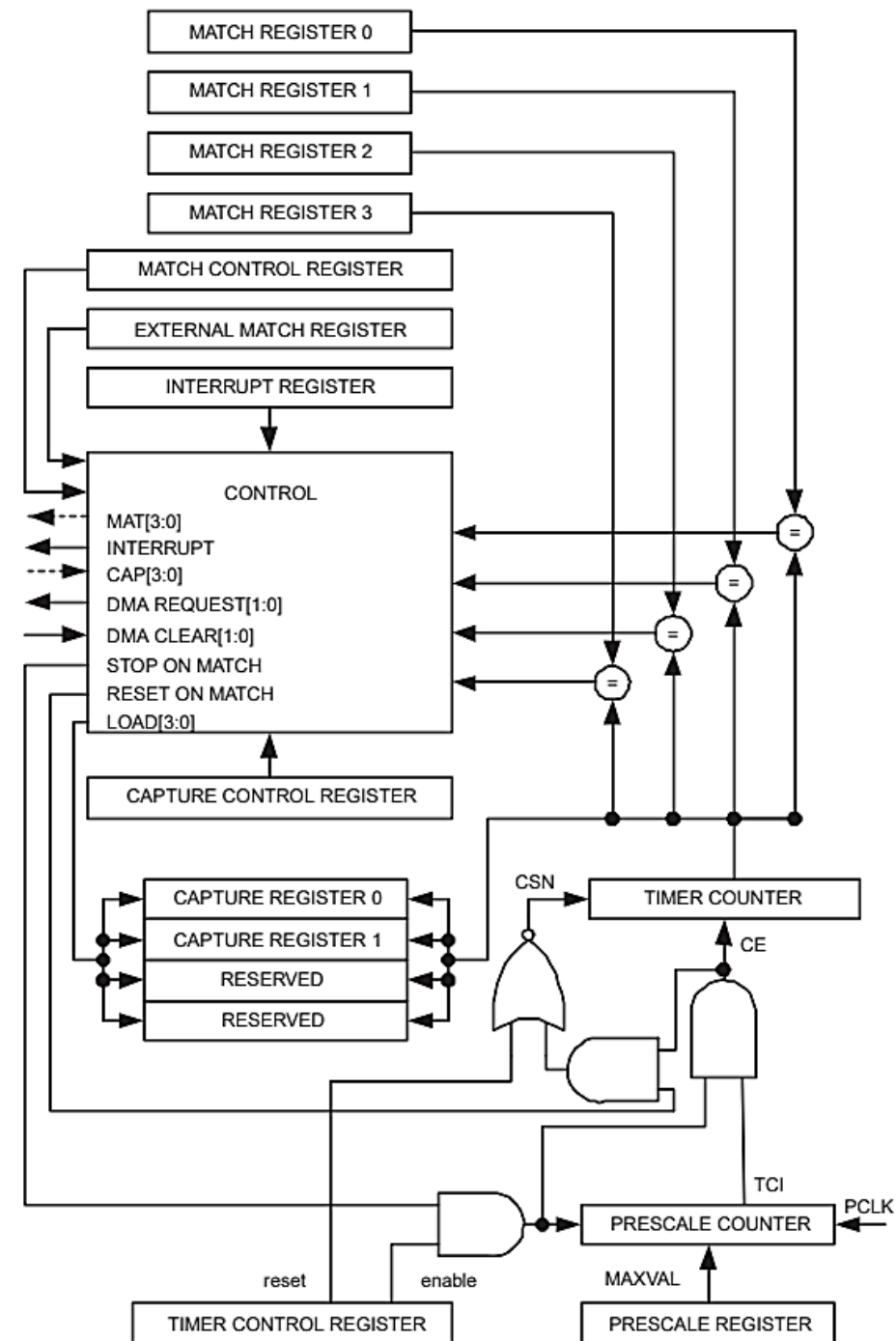
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Módulo Timer

Los timers de 32bits (Timer0/1/2/3) de la LPC1769 pueden ser utilizados en **modo temporizador (MATCH)** o en **modo contador de tiempo entre eventos (CAPTURE)**.

Aplicaciones típicas

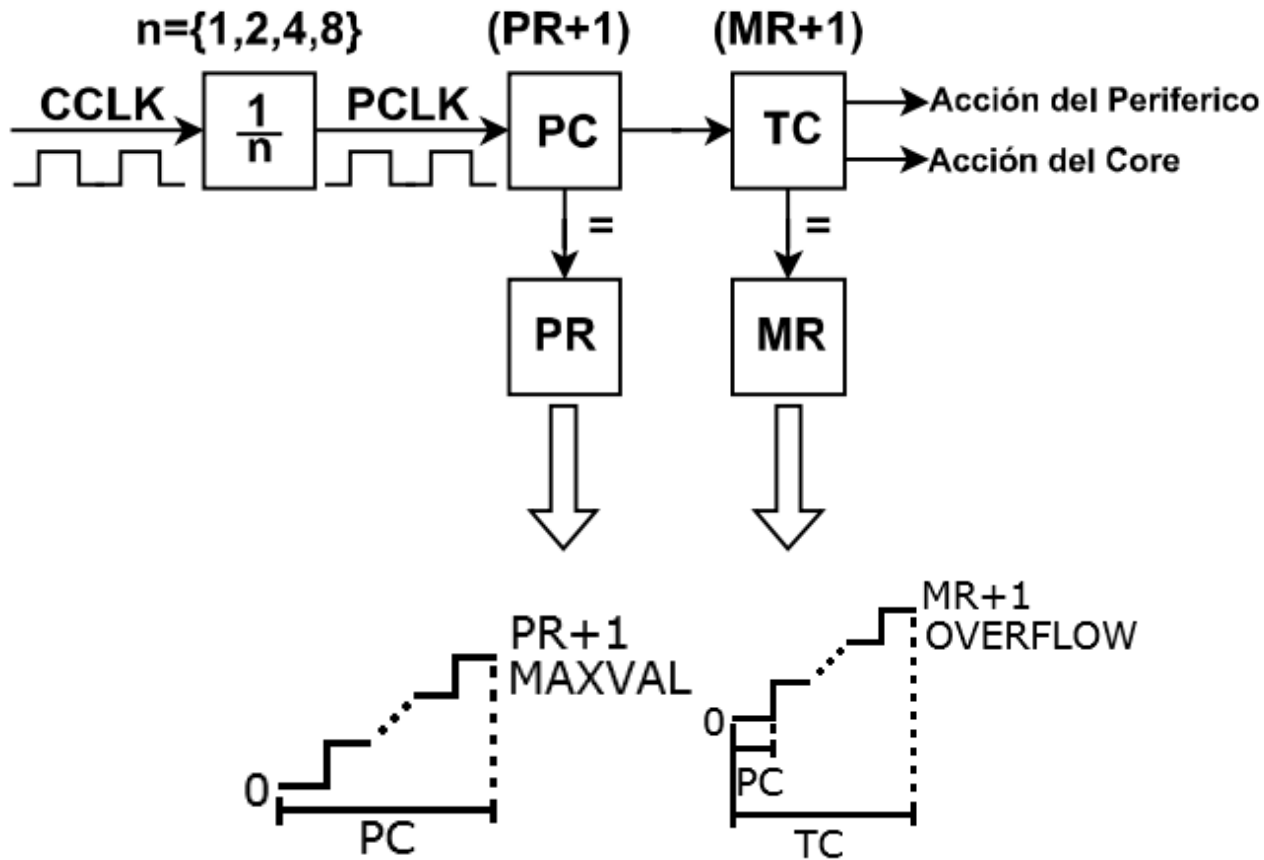
- Generación de señales periódicas o pulsos.
- Medición de frecuencias y períodos.
- Implementación de relojes en tiempo real de alta precisión (cuando no se requiere RTC).
- Captura de eventos rápidos en sistemas embebidos.
- Sincronización de procesos en aplicaciones de control.



Módulo Timer: Modo MATCH

Inicialmente, la señal de reloj CCLK ingresa a un divisor de frecuencia externo (PCLKSEL0) entregando la señal de reloj PCLK al periférico. Esta señal comanda al registro **PC (Prescale Counter)** que se compara con el valor del registro **PR (Prescale Register)**, y cuando se igualan, se reinicia a cero el registro PC y se incrementa en uno el registro **TC (Timer Counter)**. Por tanto, se produce una cuenta ascendente. A su vez, el registro TC se compara con los registros **MR[0/1/2/3] (Match Register 0/1/2/3)**, y cuando se igualan se produce alguno de los siguientes eventos de Match:

- **Acción del Periférico:** Generar señal de salida en un pin mediante el registro EMR (External Match Register)
 - Pin HIGH
 - Pin LOW
 - Pin TOGGLE (HIGH & LOW)
 - Pin NOTHING
- **Acción del Core:** Generar Interrupción mediante el NVIC



Módulo Timer: Modo MATCH

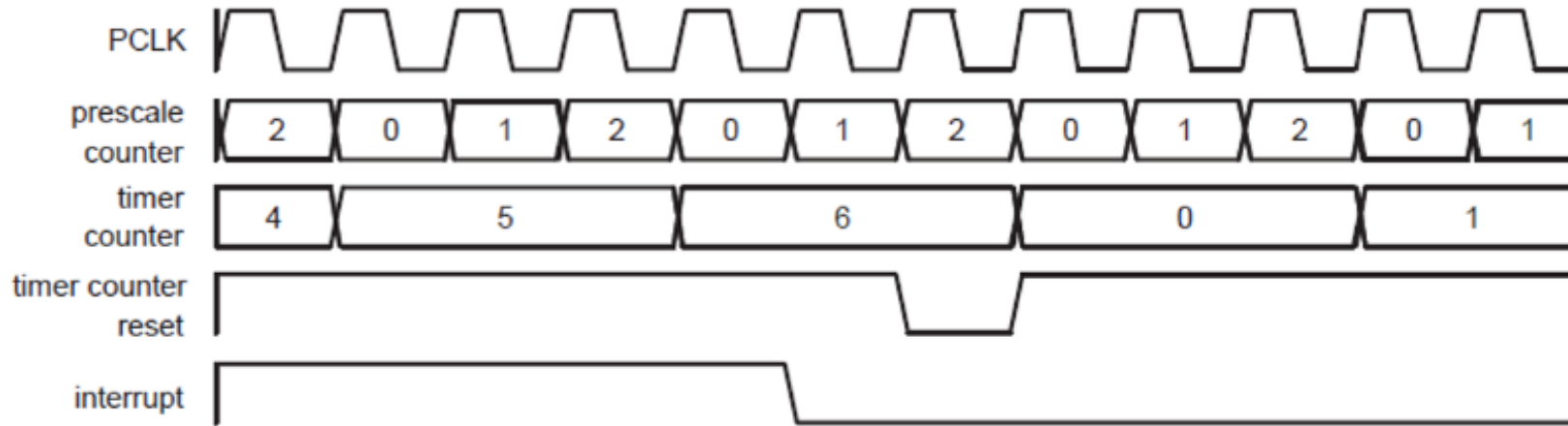


Fig 115. A timer cycle in which PR=2, MRx=6, and both interrupt and reset on match are enabled.

El valor de referencia para el MATCHx se define como:

$$t_{match} = \frac{1}{\left(\frac{CCLK}{n}\right)} (PR + 1)(MR + 1) \Rightarrow MR = \left\lceil t_{match} \left(\frac{CCLK}{n}\right) \frac{1}{(PR + 1)} \right\rceil - 1$$

Donde n es el valor del divisor de frecuencia externo dado por PCLKSEL0.

Módulo Timer: Inicialización del Timer

1) Se debe energizar el periférico a partir de los bits del registro PCON:

Power Control for Peripherals register (PCONP - address 0x400F C0C4) bit description

Bit	Symbol	Description	Reset value
1	PCTIM0	Timer/Counter 0 power/clock control bit.	1
2	PCTIM1	Timer/Counter 1 power/clock control bit.	1
22	PCTIM2	Timer 2 power/clock control bit.	0
23	PCTIM3	Timer 3 power/clock control bit.	0

Remark: On reset, Timer0/1 are enabled (PCTIM0/1 = 1), and Timer2/3 are disabled (PCTIM2/3 = 0).

Módulo Timer: Inicialización del Timer

2) Se debe seleccionar si la señal de reloj del periférico tendrá una división de frecuencia:

Peripheral Clock Selection register 0 (PCLKSEL0 - address 0x400F C1A8) bit description

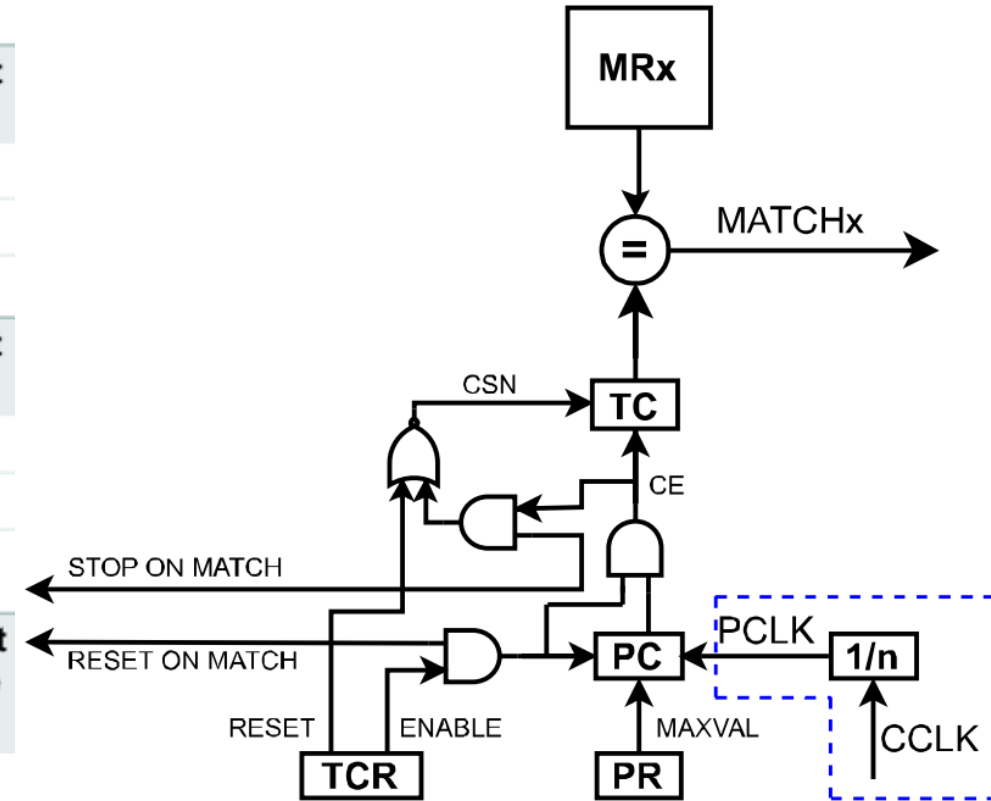
Bit	Symbol	Description	Reset value
3:2	PCLK_TIMER0	Peripheral clock selection for TIMER0.	00
5:4	PCLK_TIMER1	Peripheral clock selection for TIMER1.	00

Peripheral Clock Selection register 1 (PCLKSEL1 - address 0x400F C1AC) bit description

Bit	Symbol	Description	Reset value
13:12	PCLK_TIMER2	Peripheral clock selection for TIMER2.	00
15:14	PCLK_TIMER3	Peripheral clock selection for TIMER3.	00

Peripheral Clock Selection register bit values

PCLKSEL0 and PCLKSEL1 individual peripheral's clock select options	Function	Reset value
00	PCLK_peripheral = CCLK/4	00
01	PCLK_peripheral = CCLK	
10	PCLK_peripheral = CCLK/2	
11	PCLK_peripheral = CCLK/8, except for CAN1, CAN2, and CAN filtering when "11" selects = CCLK/6.	



Módulo Timer: Inicialización del Timer

3) Se debe seleccionar la modalidad de temporizador o contador del periférico a partir del registro CTCR:

Count Control Register (T[0/1/2/3]CTCR - addresses 0x4000 4070, 0x4000 8070, 0x4009 0070, 0x4009 4070) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	Counter/Timer Mode		This field selects which rising PCLK edges can increment the Timer's Prescale Counter (PC), or clear the PC and increment the Timer Counter (TC).	00
		00	Timer Mode: the TC is incremented when the Prescale Counter matches the Prescale Register. The Prescale Counter is incremented on every rising PCLK edge.	
		01	Counter Mode: TC is incremented on rising edges on the CAP input selected by bits 3:2.	
		10	Counter Mode: TC is incremented on falling edges on the CAP input selected by bits 3:2.	
		11	Counter Mode: TC is incremented on both edges on the CAP input selected by bits 3:2.	
3:2	Count Input Select		When bits 1:0 in this register are not 00, these bits select which CAP pin is sampled for clocking.	00
		00	CAPn.0 for TIMERN	
		01	CAPn.1 for TIMERN	
		10	Reserved	
		11	Reserved	
Note: If Counter mode is selected for a particular CAPn input in the TnCTCR, the 3 bits for that input in the Capture Control Register (TnCCR) must be programmed as 000. However, capture and/or interrupt can be selected for the other 3 CAPn inputs in the same timer.				
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Módulo Timer: Inicialización del Timer

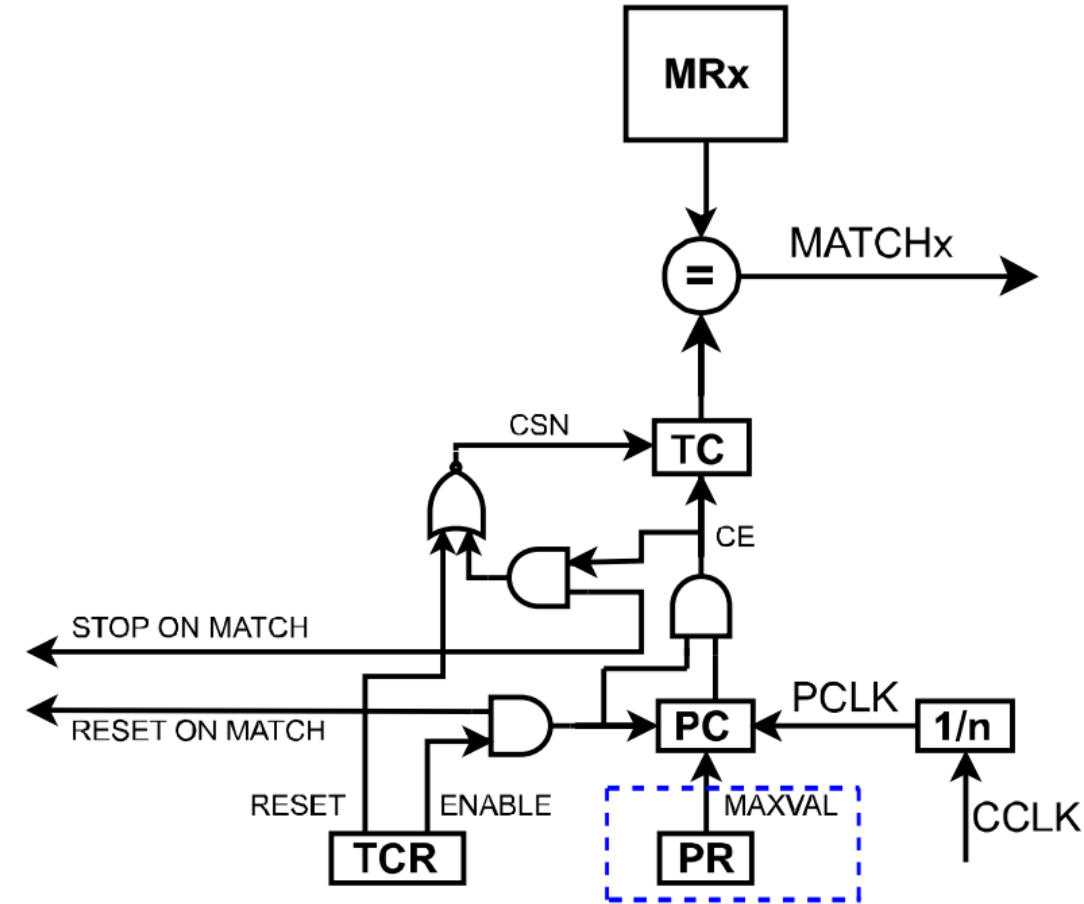
4) Se debe configurar el valor máximo que tendrá el registro PR:

Prescale register (T0PR - T3PR, 0x4000 400C, 0x4000 800C, 0x4009 000C, 0x4009 400C)

The 32-bit Prescale register specifies the maximum value for the Prescale Counter.

$$PR = PCLK_{TIMER0}(\text{Time}_{\text{Overflow}_{PC}} / 1 \times 10^6) - 1$$

Donde $\text{Time}_{\text{Overflow}_{PC}}$ es la base de tiempo de desborde del Prescaler en $[\mu s]$.



Módulo Timer: Inicialización del Timer

5) Se debe reiniciar el periférico:

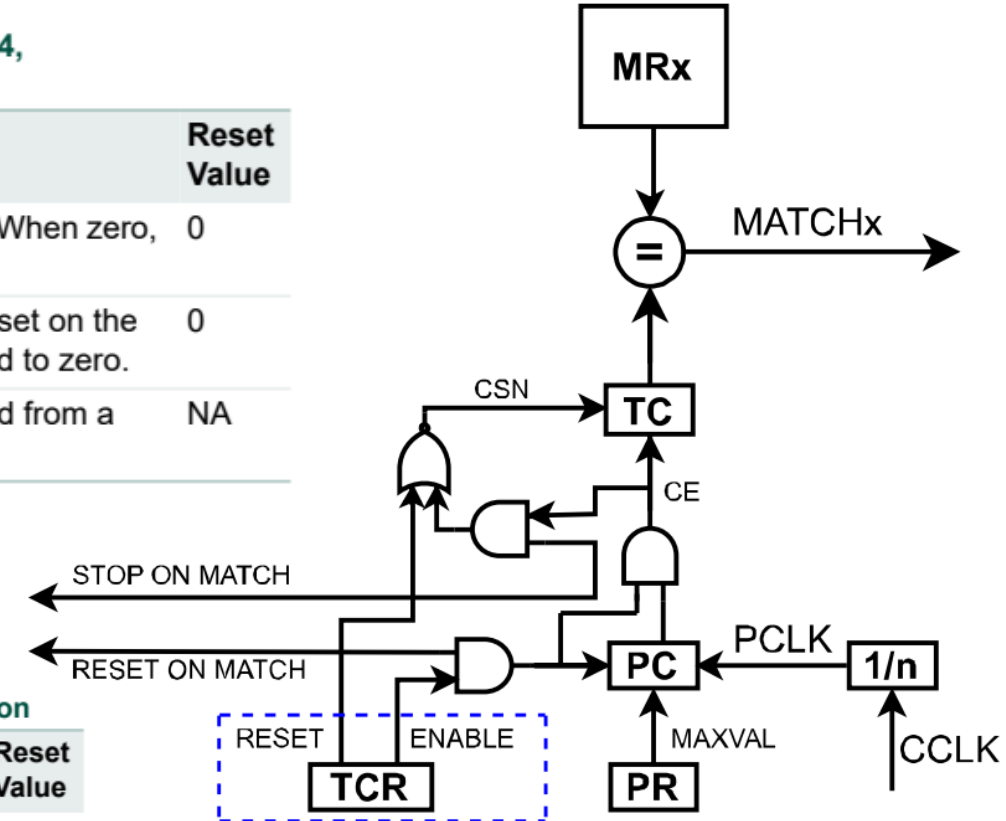
Timer Control Register (TCR, TIMERN: TnTCR - addresses 0x4000 4004, 0x4000 8004, 0x4009 0004, 0x4009 4004) bit description

Bit	Symbol	Description	Reset Value
0	Counter Enable	When one, the Timer Counter and Prescale Counter are enabled for counting. When zero, the counters are disabled.	0
1	Counter Reset	When one, the Timer Counter and the Prescale Counter are synchronously reset on the next positive edge of PCLK. The counters remain reset until TCR[1] is returned to zero.	0
31:2	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

6) Se debe limpiar la bandera de interrupción por MATCH (MRx) o por CAPTURE (CRx):

Interrupt Register (T[0/1/2/3]IR - addresses 0x4000 4000, 0x4000 8000, 0x4009 0000, 0x4009 4000) bit description

Bit	Symbol	Description	Reset Value
0	MR0 Interrupt	Interrupt flag for match channel 0.	0
1	MR1 Interrupt	Interrupt flag for match channel 1.	0
2	MR2 Interrupt	Interrupt flag for match channel 2.	0
3	MR3 Interrupt	Interrupt flag for match channel 3.	0
4	CR0 Interrupt	Interrupt flag for capture channel 0 event.	0
5	CR1 Interrupt	Interrupt flag for capture channel 1 event.	0
31:6	-	Reserved	-

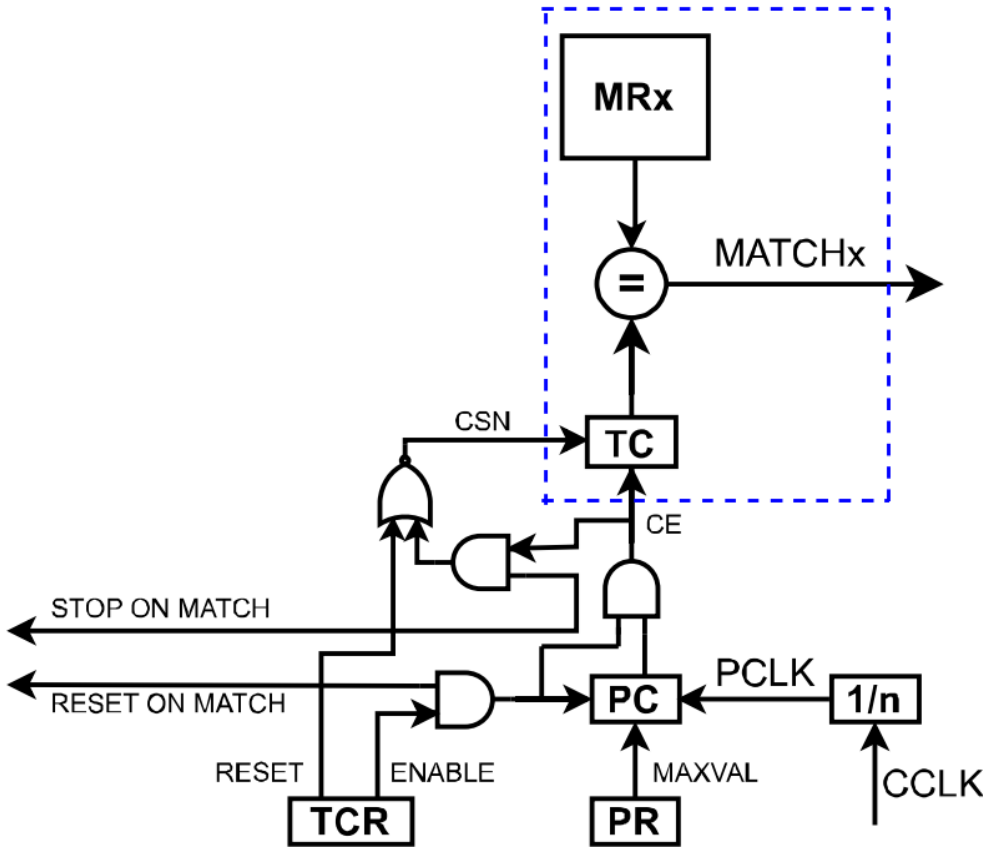


Módulo Timer: Configuración del Timer en Modo Match

1) Se debe establecer el valor máximo del registro TC a partir de los registros MRx:

TIMER/COUNTER0-3 register map

Generic Name	Description	Access	Reset Value ^[1]
MR0	Match Register 0. MR0 can be enabled through the MCR to reset the TC, stop both the TC and PC, and/or generate an interrupt every time MR0 matches the TC.	R/W	0
MR1	Match Register 1. See MR0 description.	R/W	0
MR2	Match Register 2. See MR0 description.	R/W	0
MR3	Match Register 3. See MR0 description.	R/W	0



Módulo Timer: Configuración del Timer en Modo Match

2) Se debe configurar la opción de interrupción por MATCH, la opción de habilitación de reset por MATCH, y la opción de stop por MATCH:

Match Control Register (T[0/1/2/3]MCR - addresses 0x4000 4014, 0x4000 8014, 0x4009 0014, 0x4009 4014) bit description

Bit	Symbol	Value	Description	Reset Value	Bit	Symbol	Value	Description	Reset Value
0	MR0I	1	Interrupt on MR0: an interrupt is generated when MR0 matches the value in the TC.	0	6	MR2I	1	Interrupt on MR2: an interrupt is generated when MR2 matches the value in the TC.	0
		0	This interrupt is disabled				0	This interrupt is disabled	
1	MR0R	1	Reset on MR0: the TC will be reset if MR0 matches it.	0	7	MR2R	1	Reset on MR2: the TC will be reset if MR2 matches it.	0
		0	Feature disabled.				0	Feature disabled.	
2	MR0S	1	Stop on MR0: the TC and PC will be stopped and TCR[0] will be set to 0 if MR0 matches the TC.	0	8	MR2S	1	Stop on MR2: the TC and PC will be stopped and TCR[0] will be set to 0 if MR2 matches the TC.	0
		0	Feature disabled.				0	Feature disabled.	
3	MR1I	1	Interrupt on MR1: an interrupt is generated when MR1 matches the value in the TC.	0	9	MR3I	1	Interrupt on MR3: an interrupt is generated when MR3 matches the value in the TC.	0
		0	This interrupt is disabled				0	This interrupt is disabled	
4	MR1R	1	Reset on MR1: the TC will be reset if MR1 matches it.	0	10	MR3R	1	Reset on MR3: the TC will be reset if MR3 matches it.	0
		0	Feature disabled.				0	Feature disabled.	
5	MR1S	1	Stop on MR1: the TC and PC will be stopped and TCR[0] will be set to 0 if MR1 matches the TC.	0	11	MR3S	1	Stop on MR3: the TC and PC will be stopped and TCR[0] will be set to 0 if MR3 matches the TC.	0
		0	Feature disabled.				0	Feature disabled.	
					31:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.		NA

Módulo Timer: Configuración del Timer en Modo Match

3) Se debe configurar el evento a producirse luego de que ocurra una interrupción por MATCH a partir del registro EMR:

External Match Register (T[0/1/2/3]EMR - addresses 0x4000 403C, 0x4000 803C, 0x4009 003C, 0x4009 403C) bit description

Bit	Symbol	Description	Reset Value
0	EM0	External Match 0. When a match occurs between the TC and MR0, this bit can either toggle, go low, go high, or do nothing, depending on bits 5:4 of this register. This bit can be driven onto a MATn.0 pin, in a positive-logic manner (0 = low, 1 = high).	0
1	EM1	External Match 1. When a match occurs between the TC and MR1, this bit can either toggle, go low, go high, or do nothing, depending on bits 7:6 of this register. This bit can be driven onto a MATn.1 pin, in a positive-logic manner (0 = low, 1 = high).	0
2	EM2	External Match 2. When a match occurs between the TC and MR2, this bit can either toggle, go low, go high, or do nothing, depending on bits 9:8 of this register. This bit can be driven onto a MATn.2 pin, in a positive-logic manner (0 = low, 1 = high).	0
3	EM3	External Match 3. When a match occurs between the TC and MR3, this bit can either toggle, go low, go high, or do nothing, depending on bits 11:10 of this register. This bit can be driven onto a MATn.3 pin, in a positive-logic manner (0 = low, 1 = high).	0
5:4	EMC0	External Match Control 0. Determines the functionality of External Match 0. Table 433 shows the encoding of these bits.	00
7:6	EMC1	External Match Control 1. Determines the functionality of External Match 1. Table 433 shows the encoding of these bits.	00
9:8	EMC2	External Match Control 2. Determines the functionality of External Match 2. Table 433 shows the encoding of these bits.	00
11:10	EMC3	External Match Control 3. Determines the functionality of External Match 3. Table 433 shows the encoding of these bits.	00
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

External Match Control

EMR[11:10], EMR[9:8], EMR[7:6], or EMR[5:4]	Function
00	Do Nothing.
01	Clear the corresponding External Match bit/output to 0 (MATn.m pin is LOW if pinned out).
10	Set the corresponding External Match bit/output to 1 (MATn.m pin is HIGH if pinned out).
11	Toggle the corresponding External Match bit/output.

Módulo Timer: Modo CAPTURE

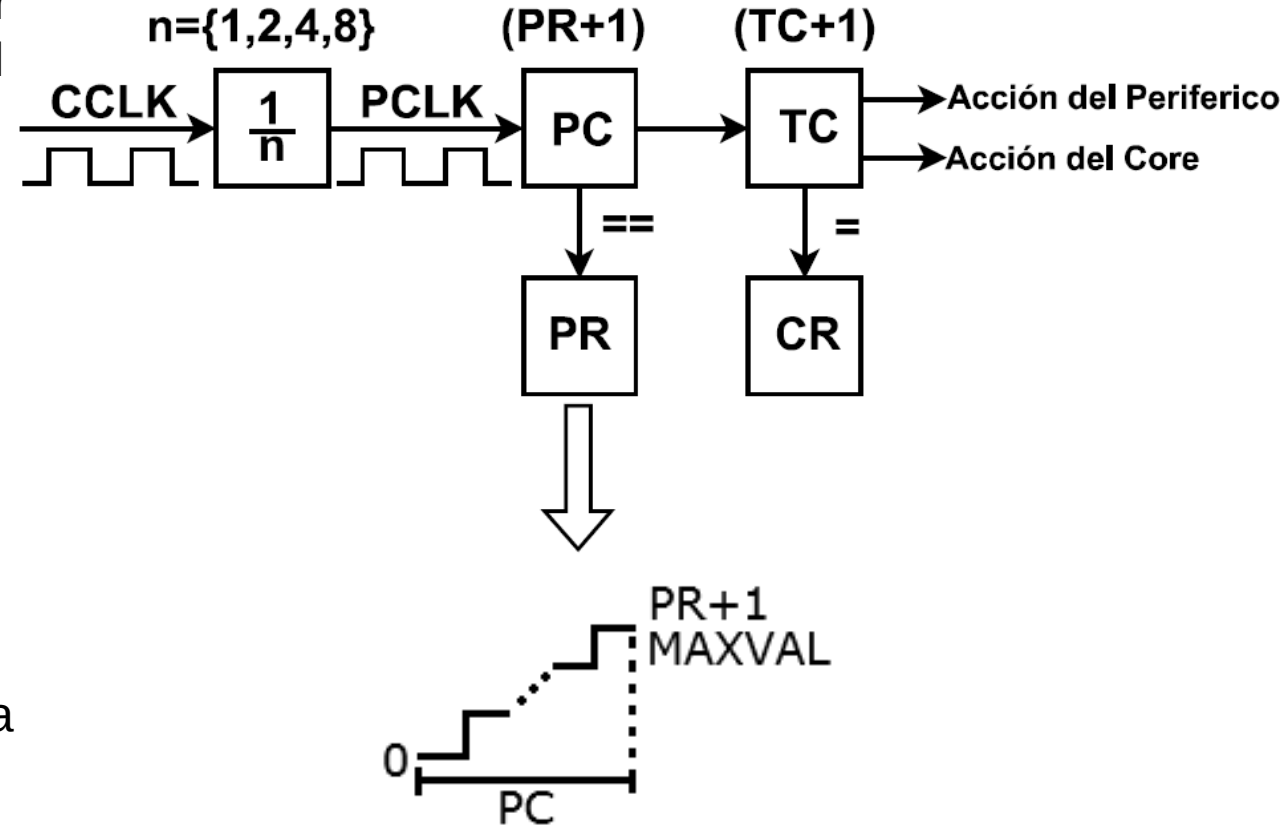
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El tiempo correspondiente a un tick del TC se determina como:

$$t_{capture} = \frac{1}{\left(\frac{CCLK}{n}\right)} (PR + 1)$$

Entonces, el tiempo entre un evento de captura será:

$$t_{delay} = t_{capture} CR$$



Nota:

Match: evento generado por comparación **interna** ($TC == MR$).

Capture: evento generado por **señal externa**, que guarda el valor de **TC**.